COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	06/10/2026	**DATE SUBMITTED:**	15/10/2026
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ACTIVITY NO. 11 **Containerization**

1. PROCEDURE OUTPUT

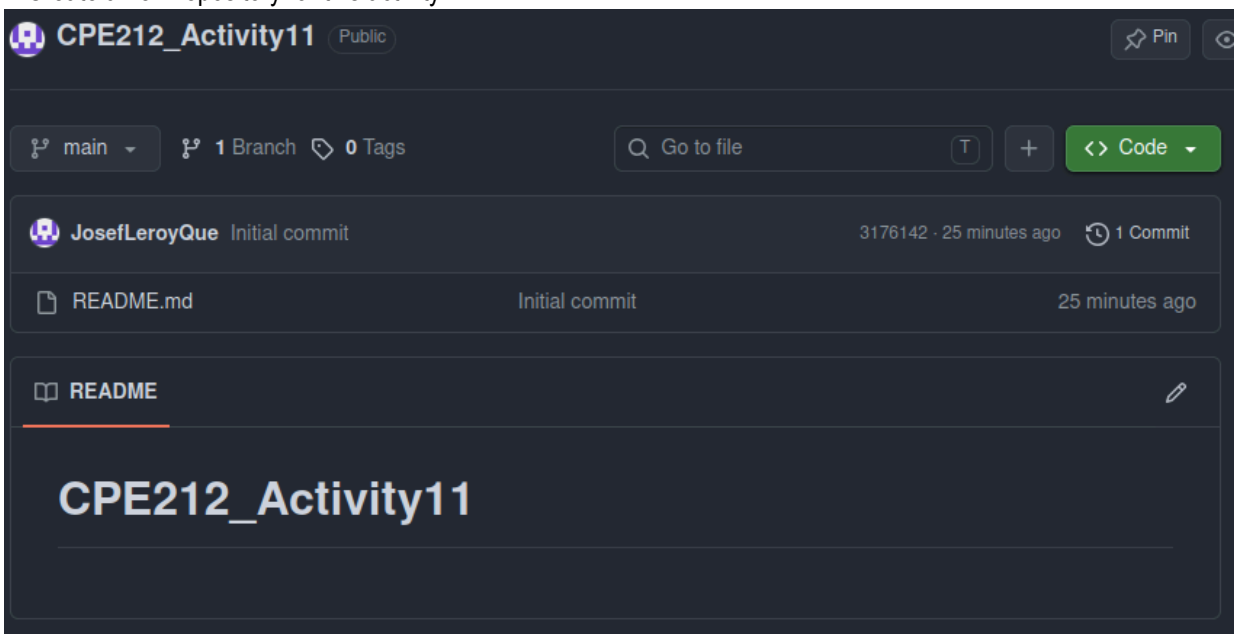
inventory.ini

```
  GNU nano 6.2                          inventory.ini
[ubuntu_nodes]
ubuntu_node1 ansible_host=192.168.56.122 ansible_user=queleroyubuntu3

[centos_nodes]
centos_node1 ansible_host=192.168.56.123 ansible_user=leroyq

[all:vars]
ansible_ssh_private_key_file=~/.ssh/id_rsa
```

1. Create a new repository for this activity.



Analysis: The new repository were created solely for this activity.

2. Install Docker and enable the docker socket.

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ sudo apt update
Hit:1 http://ph.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
```

```
 GNU nano 6.2                                        docker-installation.yml
- name: Install Docker and start the service
  hosts: all
  become: yes

  tasks:
    - name: Install Docker on Ubuntu
      ansible.builtin.apt:
        name: docker.io
        state: present
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: Add Docker repository on CentOS
      ansible.builtin.command:
        cmd: dnf config-manager --add-repo https://download.docker.com/linux/centos/docker-ce.repo
        creates: /etc/yum.repos.d/docker-ce.repo
      when: ansible_distribution == "CentOS"

    - name: Install Docker on CentOS
      ansible.builtin.dnf:
        name:
          - docker-ce
          - docker-ce-cli
          - containerd.io
        state: present
      when: ansible_distribution == "CentOS"

    - name: Start and enable Docker
      ansible.builtin.service:
        name: docker
        state: started
        enabled: yes
```

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ ansible-playbook -i inventory.ini docker-installation.yml -K
BECOME password:

PLAY [Install Docker and start the service] ************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [ubuntu_node1]
[DEPRECATION WARNING]: Distribution centos 9 on host centos_node1 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
 for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [centos_node1]

TASK [Install Docker on Ubuntu] ************************************************************************************************
skipping: [centos_node1]
ok: [ubuntu_node1]

TASK [Add Docker repository on CentOS] *****************************************************************************************
skipping: [ubuntu_node1]
ok: [centos_node1]

TASK [Install Docker on CentOS] ************************************************************************************************
skipping: [ubuntu_node1]
ok: [centos_node1]

TASK [Start and enable Docker] *************************************************************************************************
ok: [ubuntu_node1]
ok: [centos_node1]

PLAY RECAP *********************************************************************************************************************
centos_node1               : ok=4    changed=0    unreachable=0    failed=0    skipped=1    rescued=0    ignored=0
ubuntu_node1               : ok=3    changed=0    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Analysis: The docker installation as well as enabling it were successful for both ubuntu and centos nodes.

3. Add a Docker group to your current user.

```
  GNU nano 6.2                                                docker-group.yml
---
- name: Add user to Docker group
  hosts: all
  become: yes

  tasks:
    - name: Create Docker group
      ansible.builtin.group:
        name: docker
        state: present

    - name: Add current user to Docker group
      ansible.builtin.user:
        name: queleroyubuntu3
        groups: docker
        append: yes
```

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ ansible-playbook -i inventory.ini docker-group.yml -K
BECOME password:

PLAY [Add user to Docker group] ************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************
ok: [ubuntu_node1]
[DEPRECATION WARNING]: Distribution centos 9 on host centos_node1 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
 for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [centos_node1]

TASK [Create Docker group] *****************************************************************************************************
ok: [ubuntu_node1]
ok: [centos_node1]

TASK [Add current user to Docker group] ****************************************************************************************
changed: [ubuntu_node1]
changed: [centos_node1]

PLAY RECAP *********************************************************************************************************************
centos_node1               : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_node1               : ok=3    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

Analysis: Through ansible playbook, we saw that we have successfully created the docker group as well as add the current user to it.

4. Create a Dockerfile to install web and DB servers.

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ cat Dockerfile
FROM ubuntu:22.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt-get update && apt-get install -y \
    apache2 \
    mariadb-server \
    && apt-get clean \
    && rm -rf /var/lib/apt/lists/*

COPY index.html /var/www/html/index.html

EXPOSE 80 3306

CMD ["sh", "-c", "service mariadb start && exec apachectl -D FOREGROUND"]
```

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ cat index.html
<!DOCTYPE html>
<html>
<head>
    <title>CPE212 Activity 11</title>
</head>
<body>
    <h1>Docker Containerization</h1>
    <p>Welcome to our CPE212 Activity 11 project!</p>
    <p>Apache Web Server is running inside a Docker container.</p>
    <p>MariaDB is installed as the database server.</p>
</body>
</html>
```

Analysis: The dockerfile as well as the index.html were created.

5. Install and build the Dockerfile using Ansible.

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ ansible-playbook -i inventory.ini docker-build.yml -K
BECOME password:

PLAY [Build Docker image] ******************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************
ok: [ubuntu_node1]
[DEPRECATION WARNING]: Distribution centos 9 on host centos_node1 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
 for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
ok: [centos_node1]

TASK [Create project directory] ************************************************************************************************************
ok: [ubuntu_node1]
changed: [centos_node1]

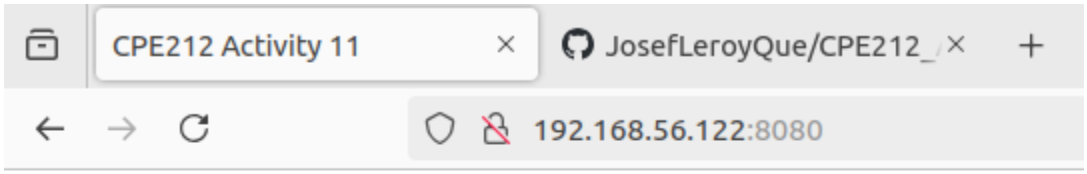
TASK [Copy Dockerfile] *********************************************************************************************************************
ok: [ubuntu_node1]
changed: [centos_node1]

TASK [Copy web page] ***********************************************************************************************************************
ok: [ubuntu_node1]
changed: [centos_node1]

TASK [Build Docker image] ******************************************************************************************************************
changed: [ubuntu_node1]
changed: [centos_node1]

PLAY RECAP *********************************************************************************************************************************
centos_node1               : ok=5    changed=4    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu_node1               : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
queleroyubuntu3@workstation:~/CPE212_Activity11$ ansible all -i inventory.ini -b -m command -a "docker images"
BECOME password:
ubuntu_node1 | CHANGED | rc=0 >>
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
cpe212-web-db:latest   9ebd00c4288c       616MB          116MB   U
ubuntu:22.04           5ec03bb3441e       119MB         31.7MB        WARNING: This output is designed for human readability. For m
achine-readable output, please use --format.
[DEPRECATION WARNING]: Distribution centos 9 on host centos_node1 should use /usr/libexec/platform-python, but is using
/usr/bin/python for backward compatibility with prior Ansible releases. A future Ansible release will default to using the
discovered platform python for this host. See https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html
 for more information. This feature will be removed in version 2.12. Deprecation warnings can be disabled by setting
deprecation_warnings=False in ansible.cfg.
centos_node1 | CHANGED | rc=0 >>
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
cpe212-web-db:latest   4f6134eb4a2c       611MB          116MB
```



Docker Containerization

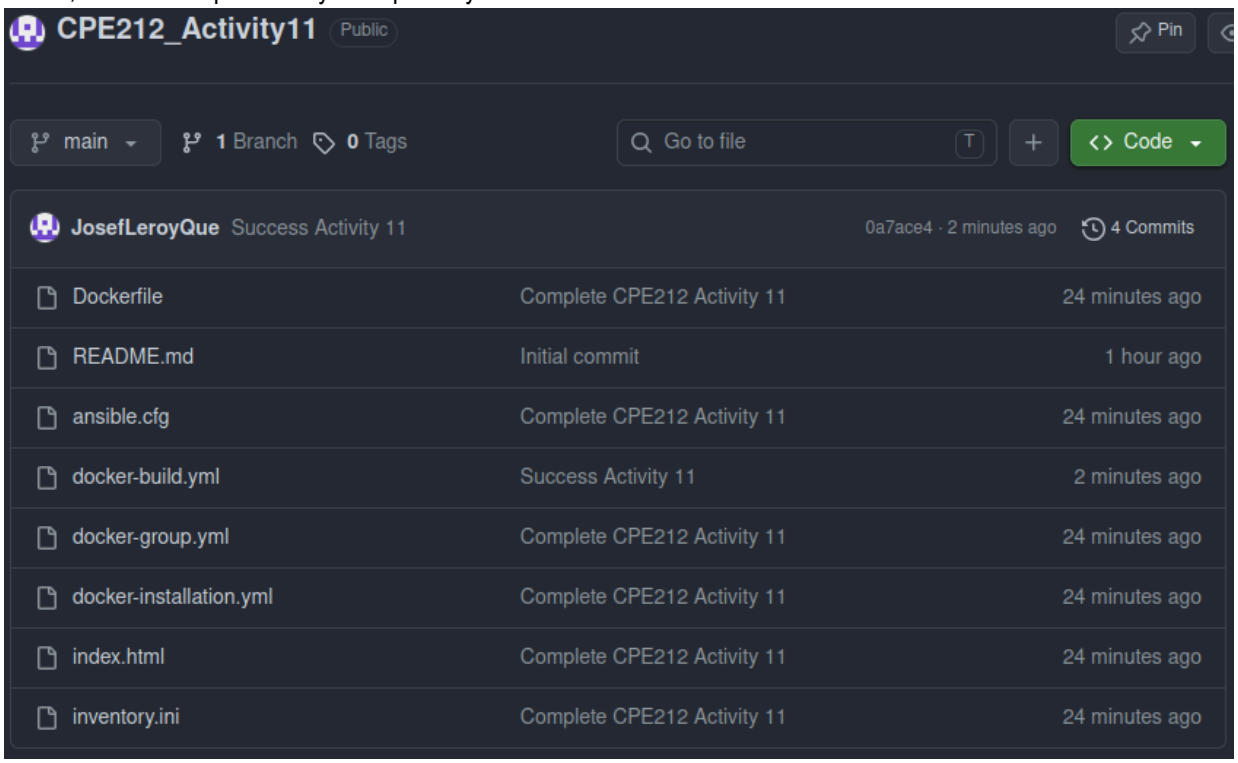
Welcome to our CPE212 Activity 11 project!

Apache Web Server is running inside a Docker container.

MariaDB is installed as the database server.

Analysis: Here, we can see the successful implementation through browser verification.

6. Add, commit and push it to your repository.



Analysis: Files were committed to github successfully.

Reflections:
Answer the following:
1. What are the benefits of implementing containerizations?
- Containerization packages all the needed things (files, codes, libraries, and etc.) of a computer program into a simple container. This solves the problem for portability like having to run a program to another PC. There are other benefits like 'efficiency' in which it is lightweight and fast, 'isolation' in which the containers are private and doesn't get affected by other containers.

2. CONCLUSION

For this laboratory activity, I was able to learn how to perform packaging of web server along with the database server of the application in Docker container using the Dockerfile prepared for this purpose. Also, I was able to learn how Ansible is used to automate the various processes of building the application itself as well as the installation process of the software so that for the completion of deployment procedure an automated process can be used. Besides of this, I had an opportunity to learn how to manage Docker services and store the project files using GitHub.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Gemini AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **AI was solely used for explaining unfamiliar concepts**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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